VARUN V AMIN

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Career Objective

Electronics and Communication Engineering student with hands-on experience in Embedded Systems, Python development, and signal processing. Skilled in PSoC microcontrollers, SystemVerilog, MATLAB, and FastAPI-based backend development. Experience engineering AI/ML pipelines during an internship at Invent Health and working with power distribution systems at MESCOM. Seeking opportunities in Embedded Systems, AI/ML applications, or core electronics engineering.

Education

NMAM Institute of Technology, Nitte — 2023 – 2027
B.Tech in Electronics and Communication Engineering

St. Aloysius PU College, Mangalore — 2021 – 2023
Pre-University – 91.6%

Holy Family English Medium High School, Surathkal — 2021
SSLC – 89.9%

Internship

AI & NLP Intern — Feb 2026 – May 2026
Invent Health, Bengaluru

- Engineered "Epsilon," a Retrieval-Augmented Generation (RAG) study suite featuring a Streamlit web portal and a React/Next.js/Electron desktop client for real-time document intelligence.
- Built end-to-end local data ingestion pipelines utilizing PyMuPDF, LangChain, and FAISS for sub-second semantic retrieval.
- Implemented a multi-model routing system via the Groq API (Llama 4, 3.3, 3.1) and fine-tuned large language models using Unsloth and LoRA adapters.
- Evaluated OCR pipelines using NVIDIA Nemotron and integrated continuous real-time audio transcription via Web Speech APIs.

Electrical Distribution Intern — Jun – Jul 2025
Mangalore Electricity Supply Company Ltd. (MESCOM), Mangalore

- Gained practical exposure to power distribution networks, substations, transformers, and transmission systems.
- Assisted engineers in load analysis, fault identification, and preventive maintenance of electrical infrastructure.
- Observed grid monitoring operations, energy metering systems, and electrical safety protocols.

Projects

RAG AI Assistant — Python, Streamlit, LangChain, FAISS, Google Gemini API

- Developed a Retrieval-Augmented Generation (RAG) chatbot that answers user queries from uploaded PDF documents using semantic search and LLM-powered responses.
- Implemented document ingestion, text chunking, embedding generation, and FAISS-based vector storage for efficient context retrieval and knowledge extraction.
- Integrated Google Gemini API with LangChain to deliver accurate, context-aware responses, enhancing document understanding and user interaction.

Industrial Machine Health Monitoring System using ESP32 & IoT — ESP32, Embedded C, Thingspeak, Arduino IDE

- Developed an IoT-based industrial machine monitoring system using ESP32 with temperature, vibration, and current sensors for real-time health assessment.
- Integrated Wi-Fi communication and ThingSpeak cloud platform for remote data visualization and continuous machine monitoring.
- Implemented threshold-based anomaly detection and alert generation to support predictive maintenance and reduce unplanned downtime.

Brain Tumor Detection Using Machine Learning — Python, OpenCV, Scikit-learn, Google Colab

- Built a brain tumor classification system using MRI images, implementing SVM, KNN, and Random Forest with a complete preprocessing pipeline.
- Applied PCA for dimensionality reduction and evaluated models using Accuracy, Precision, Recall, and F1-Score achieving best accuracy with Random forest.
- Developed a majority voting mechanism across all three classifiers for robust final prediction on new MRI images.

Science Tutor Bot — Python, Flask, LLMs, Groq API

- Developed a science-focused chatbot using LLMs, implementing a two-layer classification system with BART zero-shot classification and Groq API for accurate topic filtering and response generation.
- Fine-tuned a Llama-based model using LoRA (Unsloth) on the SciQ dataset and deployed the application as a Flask web app on Render, gaining experience in LLM fine-tuning and deployment.

Sequence Quest — Embedded C Project using a PSoC microcontroller

- Designed an embedded system using Embedded C on a PSoC microcontroller to validate and generate numerical sequences.
- Implemented keypad input, LCD interfacing, and UART communication for real-time sequence validation and display.

Technical Skills

Programming Languages: Python, C, Embedded C, Verilog, SystemVerilog
Tools Platforms: MATLAB, Vivado, PSoC Creator, VS Code
Domains: Signal Processing, Digital Design, Embedded systems, Power Systems
Languages: English, Hindi, Kannada, Tulu